Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/20/2006
PI Name:	Bhattacharjee, Amitava Ph.D.		
Project Title:	Theory and Simulation of Ground-Based and Microgravity Dusty Plasma Experiments		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	03824	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	02/01/2000	**End Date:**	10/05/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	note- grant extended, new NAG#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GB10G		
Performance Goal No.:			
Performance Goal Text:			
	The principal goal of this research program is to carry out analytical theory and computer simulations of dusty plasma crystals and liquids in support of the co-located, ground-based laboratory experiments of Professor J. Goree and his group at the University of Iowa as well as the microgravity experiments of the group at the Max Planck Institute for Extraterrestrial Physics (Germany) led by Professor G. Morfill, on parabolic aircraft flights, sounding rockets, and the International Space Station. The microscopic structure and dynamics of crystalline and liquid phases have been studied in the ground-based and microgravity experiments. The experimental system is a dusty plasma, consisting of highly charged monodisperse microspheres levitated in a charge-neutral laboratory plasma. The spheres interact through a shielded Coulomb potential, causing them to arrange in liquid or solid-like structures. Compared to similar colloidal crystal experiments, the dusty plasma crystal offers significant advantages, including a rapid equilibrium and direct		

Task Description:	imaging of all the particles in a 3D suspension. Ground-based as well as microgravity experiments during the first year of the NASA-supported Iowa-Germany collaboration have identified new physical phenomena that need to be understood in order to design future experiments. Of great interest is the spontaneous generation of voids in both the ground-based and microgravity experiment. These voids can inhibit the formation of high-quality crystals. A primary goal of the theory program is to develop a nonlinear theory of void formation because it will suggest ways in which the phenomenon might be controlled. Another phenomenon identified in ground-based experiments is the formation of Mach cones. These cones are predicted by theory to form in the solar system, and in particular inside the planetary rings of the planet Saturn. The measurement of wakes and opening angles for these Mach cones can potentially provide non-intrusive measurements of dusty plasma properties in Saturns rings when the Cassini spacecraft arrives in that neighborhood in 2004. A molecular dynamics simulation, using codes developed by the Iowa group, is proposed to develop an understanding of these Mach cones. During the last few years, the Iowa group has invested substantial effort in the analytical theory and simulation (using particle-in-cell as well as molecular dynamics techniques) of waves in dusty plasma crystals and liquids. These theories have helped elucidate outstanding features of ground-based observations and brought out new features that will stimulate new experimental studies. A comprehensive analytical and computational study of linear and nonlinear longitudinal and transverse waves in dusty plasma crystals, including 3D studies relevant to microgravity experiments, is proposed. To carry out theoretical modeling and computer simulation of ground-based and microgravity dusty plasma experiments. The task is expected to produce greater understanding and control of three-dimensional complex plasma crystals, their formation and dynamics under microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)